

B-convergence

Prothero-Robinson:

$$u'(t) = \lambda(u - \phi(t)) + \phi'(t) \quad \lambda < 0$$

$$u(t_0) = \phi(t_0)$$

$$\Rightarrow u(t) = \phi(t)$$

Apply a RK method to this problem:

$$Y_i = U_n + h \sum_{j=1}^s a_{ij} [\lambda(Y_j - \phi(t_n + c_j h)) + \phi'(t_n + c_j h)]$$

$$U_{n+1} = U_n + h \sum_{j=1}^s b_j [\lambda(Y_j - \phi(t_n + c_j h)) + \phi'(t_n + c_j h)]$$

Replace num. soln. with exact soln.: $U_n \rightarrow \phi(t_n)$
 $Y_i \rightarrow \phi(t_n + c_i h)$

$$\phi(t_n + c_i h) = \phi(t_n) + h \sum_j a_{ij} \phi'(t_n + c_j h) + \tau_i$$

$$\phi(t_{n+1}) = \phi(t_n) + h \sum_j b_j \phi'(t_n + c_j h) + \tau_{s+1}$$

$$\phi(t_n + c_i h) = \sum_{k=0}^{\infty} \frac{c_i^k h^k}{k!} \phi^{(k)}(t_n)$$

$$\phi'(t_n + c_i h) = \sum_{k=0}^{\infty} \frac{c_i^k h^k}{k!} \phi^{(k+1)}(t_n) = \sum_{k=1}^{\infty} \frac{c_i^{k-1} h^{k-1}}{(k-1)!} \phi^{(k)}(t_n)$$

Substitute:

$$\sum_{k=0}^{\infty} \frac{c_i^k h^k}{k!} \phi^{(k)}(t_n) = \phi(t_n) + h \sum_j a_{ij} \sum_{k=1}^{\infty} \frac{c_j^{k-1} h^{k-1}}{(k-1)!} \phi^{(k)}(t_n) + \tau_i$$

$$\sum_{k=0}^{\infty} \frac{h^k}{k!} \phi^{(k)}(t_n) = \phi(t_n) + h \sum_j b_j \sum_{k=1}^{\infty} \frac{c_j^{k-1} h^{k-1}}{(k-1)!} \phi^{(k)}(t_n) + \tau_{s+1}$$

$$\tau = \sum_{k=1}^{\infty} h^k \left[\frac{1}{k!} C^k - \frac{1}{(k-1)!} A C^{k-1} \right] \phi^{(k)}(t_n) \quad (1)$$

$$\tau = \begin{bmatrix} \tau_1 \\ \tau_2 \\ \vdots \\ \tau_s \end{bmatrix}$$

$$\tau_{s+1} = \sum_{k=1}^{\infty} h^k \left[\frac{1}{k!} - \frac{1}{(k-1)!} b^T C^{k-1} \right] \phi^{(k)}(t_n) \quad (2)$$

We have $\tau_{s+1} = \mathcal{O}(h^{p+1})$ only if

$$B(p): b^T C^{k-1} = \frac{1}{k} \text{ for } k=1, 2, \dots, p$$

We have $\tau = \mathcal{O}(h^{q+1})$ only if

$$C(q): A C^{k-1} = \frac{1}{k} C^k \text{ for } k=1, 2, \dots, q$$

We call q the order of the stages.

The minimum of q and p is called the stage order of the method.

What is the stage order of

$$\begin{array}{c|cc} 0 & 0 & 0 \\ 1 & 1 & 0 \\ \hline & \frac{1}{2} & \frac{1}{2} \end{array} ? \quad \underline{1}$$

All explicit methods have stage order 1.

Diagonally implicit methods have stage order $q \leq 2$.

$$\phi = \begin{bmatrix} \phi(t_n + c_1 h) \\ \vdots \\ \phi(t_n + c_s h) \end{bmatrix}$$

$$\mathbb{I} = \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix} \in \mathbb{R}^s$$

Global error

$$Y_i = U_n + h \sum_{j=1}^s a_{ij} \left[\lambda(Y_j - \phi(t_n + c_j h)) + \cancel{\phi'(t_n + c_j h)} \right]$$

$$- \left(\phi(t_n + c_i h) = \phi(t_n) + h \sum_j a_{ij} \cancel{\phi'(t_n + c_j h)} + \tau_i \right)$$

$$Y - \phi = (U_n - \phi(t_n)) \mathbb{1} + h \lambda A (Y - \phi) - \tau \quad z = h \lambda$$

$$(I - zA)(Y - \phi) = (U_n - \phi(t_n)) \mathbb{1} - \tau$$

$$Y - \phi = (I - zA)^{-1} [(U_n - \phi(t_n)) \mathbb{1} - \tau]$$

Now repeat for the soln. update:

$$U_{n+1} = U_n + h \sum_{j=1}^s b_j \left[\lambda(Y_j - \phi(t_n + c_j h)) + \cancel{\phi'(t_n + c_j h)} \right]$$

$$- \left(\phi(t_n + h) = \phi(t_n) + h \sum_j b_j \cancel{\phi'(t_n + c_j h)} + \tau_{s+1} \right)$$

$$U_{n+1} - \phi(t_n + h) = U_n - \phi(t_n) + \lambda h b^T (Y - \phi) - \tau_{s+1}$$

$$= U_n - \phi(t_n) + z b^T (I - zA)^{-1} [(U_n - \phi(t_n)) \mathbb{1} - \tau] - \tau_{s+1}$$

$$= \left[1 + z b^T (I - zA)^{-1} \mathbb{1} \right] (U_n - \phi(t_n)) - z b^T (I - zA)^{-1} \tau - \tau_{s+1}$$

$$R(z)$$

Setting $U_\lambda = \phi(t_h)$ and requiring order p , we have

$$z b^T (I - zA)^{-1} \tau + \tau_{s+1} = \mathcal{O}(h^{p+1})$$

$\tau_{s+1} = \mathcal{O}(h^{p+1})$: Standard order (bushy tree) conditions

$$z b^T (I - zA)^{-1} \sum_{k=1}^{\infty} h^k \left[\frac{1}{k!} C^k - \frac{1}{(k-1)!} A C^{k-1} \right] = \mathcal{O}(h^{p+1})$$

Consider 2 regimes:

(A) Non-stiff: $\lambda = \mathcal{O}(1) \Rightarrow z = \mathcal{O}(h)$

$$(I - zA)^{-1} = I + zA + z^2 A^2 + \dots = \sum_{j=0}^{\infty} z^j A^j$$

so we have

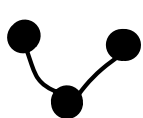
$$\lambda b^T \sum_{j=0}^{\infty} \lambda^j A^j \sum_{k=1}^{\infty} h^{k+j+1} \left[\frac{1}{k!} C^k - \frac{1}{(k-1)!} A C^{k-1} \right] = \mathcal{O}(h^{p+1})$$

These turn out to be linear combinations of the standard order conditions.

For example: $j=0, k=2$

$$\lambda b^T \cancel{h^3} \left(\frac{1}{2} C^2 - A C \right) = 0 \Rightarrow \frac{1}{2} b^T C^2 = b^T A C$$

This comes from the 3rd order conditions:



$$b^T C^2 = \frac{1}{3}$$



$$b^T A C = \frac{1}{6}$$

(B) Stiff regime: $h = \mathcal{O}\left(\frac{1}{|\lambda|}\right)$ $|z| = \mathcal{O}(1)$

So we consider $h \rightarrow 0$ while z remains large.

Again we can formally write

$$z b^T \sum_{j=0}^{\infty} z^j A^j \sum_{k=1}^{\infty} h^k \left[\frac{1}{k!} C^k - \frac{1}{(k-1)!} A C^{k-1} \right] = \mathcal{O}(h^{p+1})$$

but now this requires that

$$h^k b^T A^j (C^k - k A C^{k-1}) = \mathcal{O}(h^{p+1})$$

i.e.

$$b^T A^j (C^k - k A C^{k-1}) = 0 \quad \text{for } k=1, 2, \dots, p.$$

for all j .

So we should force

$$C^k = k A C^{k-1} \quad \text{for } 1 \leq k \leq p.$$

Thus the method must have stage order p .

Alternatively, we could hope to have

$$C^k - k A C^{k-1} \text{ orthogonal to } b^T A^j$$

for $1 \leq k \leq p$ and for all j . (Weak stage order)

Collocation Methods

An s -stage method can have stage order at most $q=s$. Such methods are fully implicit and completely determined by the abscissas.

Gauss Methods

- Gauss quadrature nodes
- $p=2s$, $q=s$
- A-stable
- Symplectic
- Not L-stable

1-stage:

$$\begin{array}{c|c} \frac{1}{2} & \frac{1}{2} \\ \hline & 1 \end{array}$$

Implicit midpoint method

Other collocation families:

- Lobatto
- Radau